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## Problem

- HDRIs are standard light sources for image-based lighting, but casual LDR panoramas **clip at the saturated area**.
- Clipped highlights destroy **diffuse** irradiance and **specular** reflections.
- Pixel losses treat all errors similarly, even though rendering quality is dominated by lighting integrals.

## Related Work

- Single-image HDR reconstruction improves local pixels, but does not directly supervise IBL behavior.
- DiffHDR shows the value of video diffusion priors for exposure recovery.

**Missing: losses aligned with how HDRIs are rendered.**

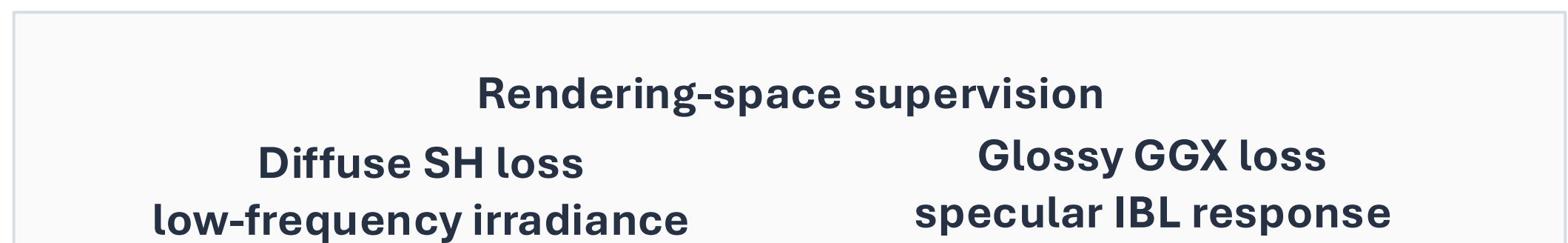
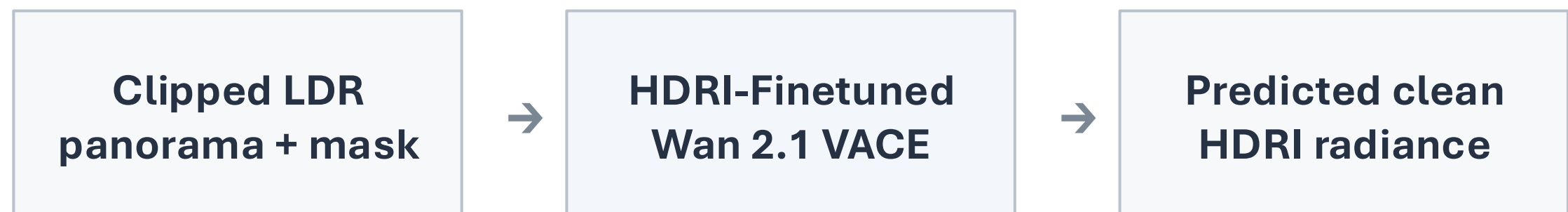
## Our Approach

- Fine-tune Wan 2.1 VACE 14B with LoRA for clipped-HDRI promotion.
- Use mask-conditioned inpainting of saturated highlight regions.
- Add diffuse SH and glossy GGX rendering-space losses for IBL fidelity.
- Train in linear radiance with log encoding for VAE compatibility.

## References

- [1] Poly Haven. 2024. <https://polyhaven.com/>
- [2] Debevec. Rendering synthetic objects into real scenes. SIGGRAPH 2008 Classes.
- [3] Eilertsen et al. HDR image reconstruction from a single exposure. TOG 2017.
- [4] Hu et al. Single-image HDR reconstruction by learning reverse camera pipeline. CVPR 2020.
- [5] Hu et al. LoRA: Low-rank adaptation of large language models. ICLR 2022.
- [6] Wan Team. Wan: Open and advanced large-scale video generative models. 2025.
- [7] Ramamoorthi and Hanrahan. Efficient irradiance environment maps. SIGGRAPH 2001.
- [8] Walter et al. Microfacet models for rough surfaces. Rendering Techniques 2007.
- [9] Yu et al. DiffHDR: Re-Exposing LDR Videos with Video Diffusion Models. 2026.

## Method

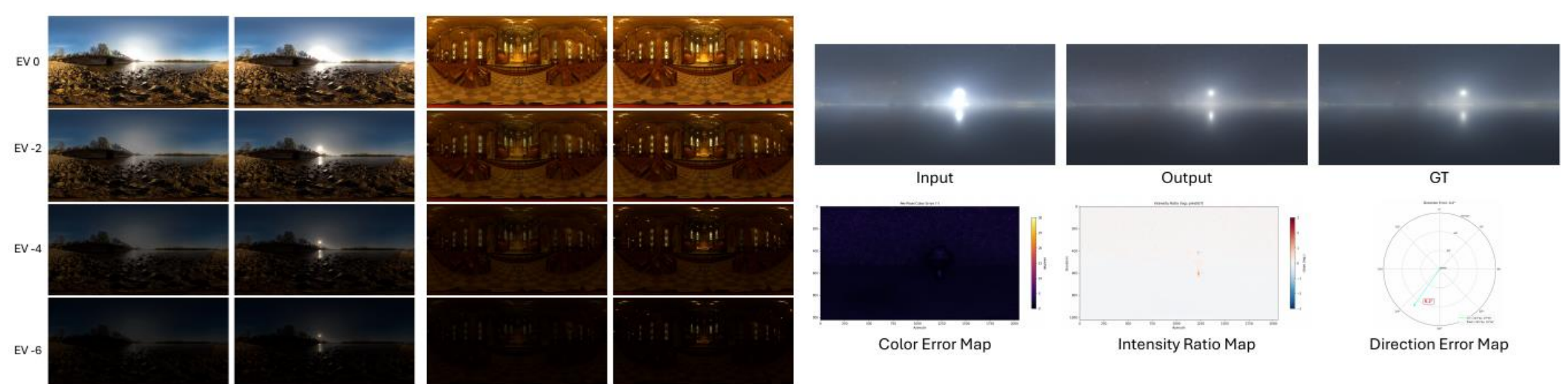


$$\mathcal{L}_{\text{diffuse}} = \frac{1}{27} \sum_{c,k} (\log(1 + c_k^{\text{pred}}) - \log(1 + c_k^{\text{gt}}))^2 \quad \mathcal{L}_{\text{glossy}} = \|\log(1 + \mathbf{E}_{\text{glossy}}^{\text{pred}}) - \log(1 + \mathbf{E}_{\text{glossy}}^{\text{gt}})\|_2^2$$

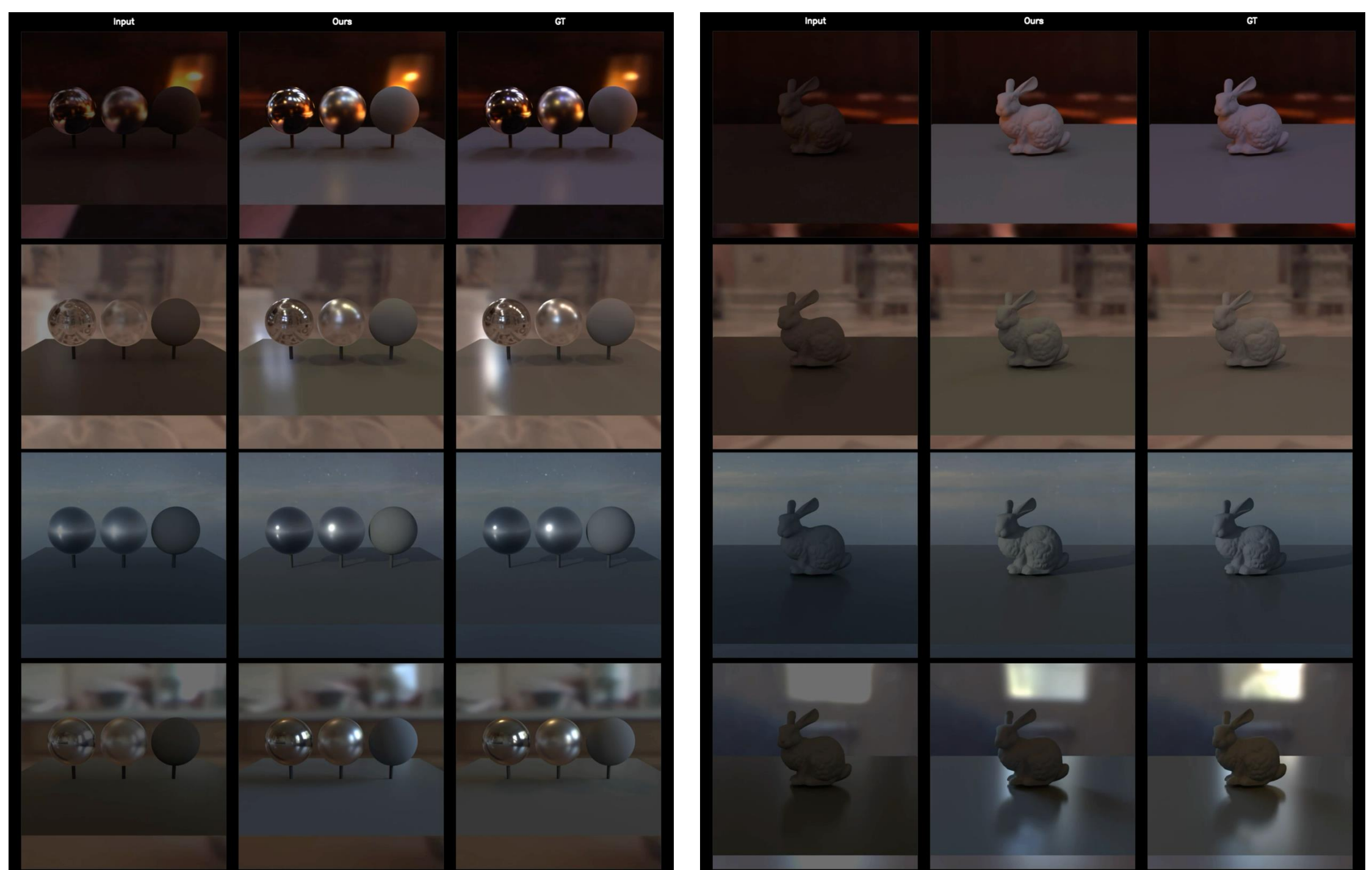
$$\mathcal{L} = \mathcal{L}_{\text{noise}} + \lambda_d \mathcal{L}_{\text{diffuse}} + \lambda_g \mathcal{L}_{\text{glossy}}, \quad \lambda_d = \lambda_g = 0.05$$

- The LDR panorama is encoded as a single-frame video input. Clipped regions are mask-conditioned for inpainting.
- Diffuse SH compares 27 second-order coefficients in log space, directly supervising irradiance.
- Glossy GGX compares prefiltered specular lobes over  $64 \times 128$  HDRI maps.

## Results



**Fig. 1** The model recovers clipped highlight details from LDR panorama input and achieve low color error, direction error, and close intensity ratio with the Ground Truth HDRI.



**Fig. 3.** Relighting comparison. Our reconstruction recovers the illumination intensity, color, shadowing, and specular reflections needed for realistic relighting.